

Targeting the novel immune checkpoint KLRG1 is markedly therapeutic against cancer through multiple lymphocyte subsets

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ABSTRACT

Background Checkpoint blockade therapies have demonstrated clinical benefit across multiple cancer types; however, many patients with immune cell infiltration remain non-responsive or develop resistance. This suggests that additional, unidentified regulatory pathways limit treatment efficacy. The purpose of this study was to investigate the role of the inhibitory receptor KLRG1 in limiting antitumor immunity and to evaluate its potential as a therapeutic target in patients refractory to first-generation checkpoint inhibitors.

Methods The study used murine models to observe KLRG1 expression following checkpoint therapy and analyzed tumor-infiltrating T cells from patients with melanoma non-responsive to anti-programmed death-1 (PD-1). We generated KLRG1 knockout mice and B16 melanoma N-cadherin (KLRG1 ligand) deletions to assess tumor growth. Additionally, a KLRG1 human knock-in mouse model and a novel anti-human KLRG1 monoclonal antibody (mAb) were developed and tested both in vitro and in vivo.

Results KLRG1 was found to be upregulated on murine CD8⁺T cells post-therapy, and high KLRG1 expression correlated with anti-PD-1 resistance in patients with melanoma. KLRG1 knockout mice showed significantly reduced tumor growth, a response dependent on CD8⁺T cells, natural killer cells, and $\gamma\delta$ T cells. Similarly, deleting N-cadherin in tumors slowed growth. The novel anti-human KLRG1 mAb enhanced human T-cell activation in vitro and significantly reduced tumor progression in humanized KLRG1 mice.

Conclusions Our findings demonstrate KLRG1 as a critical negative regulator of antitumor immunity and support KLRG1-cadherin blockade as a promising strategy for improving immunotherapy outcomes.

INTRODUCTION

Antibodies targeting programmed death-1 (PD-1) and cytotoxic T lymphocyte-associated molecule-4 (CTLA-4) have provided major clinical benefit for a variety of cancers.^{1 2} Patients who exhibit a T cell-inflamed phenotype but fail to respond to current checkpoint blockade therapy likely have other immune-regulatory pathways that

WHAT IS ALREADY KNOWN ON THIS TOPIC

⇒ Immune checkpoint blockades (eg, α -programmed death-1 (PD-1) and α -cytotoxic T lymphocyte-associated molecule-4 (CTLA-4)) are effective in many cancers, but often patients with immune cell-infiltrated tumors either do not respond or eventually become refractory, suggesting that additional regulatory pathways limit treatment efficacy. While KLRG1 is recognized as a marker of senescent/terminally differentiated T and natural killer (NK) cells, its role as a functional, druggable checkpoint receptor that limits antitumor immunity in humans has not been well established.

WHAT THIS STUDY ADDS

⇒ This study identifies KLRG1 as a critical negative regulator of antitumor immunity that is upregulated on tumor-infiltrating T-cell subsets following immunotherapy, with high expression correlating with non-response in patients with melanoma. Our work also demonstrated that the KLRG1 pathway, mediated by cadherin ligands on tumor cells, can be targeted to significantly improve tumor control in vivo involving CD8⁺T cells, $\gamma\delta$ T cells, and NK cells, as shown in a KLRG1 knockout mouse model. Furthermore, the development of a novel anti-human KLRG1 antibody effectively enhances T-cell activation in vitro and showed significant therapeutic efficacy in the human KLRG1 KI mouse model in vivo, suggesting a promising therapeutic strategy for a broad array of cancers.

HOW THIS STUDY MIGHT AFFECT RESEARCH, PRACTICE OR POLICY

⇒ Our work supports KLRG1 as a promising new therapeutic target for cancer immunotherapy. Development of anti-KLRG1 monoclonal antibodies could be a promising therapeutic strategy to overcome resistance in patients who fail to benefit from current checkpoint inhibitor therapies. Lastly, we identified correlations between cadherin expression and immunotherapy response, offering a potential biomarker to select patients for anti-KLRG1 treatment.

are functionally dominant. Our laboratory previously used gene expression profiling to characterize tumor antigen-specific CD8⁺ T cells compared with irrelevant T cells within the mouse tumor microenvironment (TME), to identify additional candidate immune checkpoints. We examined expression of these genes in the human The Cancer Genome Atlas (TCGA) database to confirm expression in T cell-inflamed human tumors. Based on this investigation, we selected the C-type lectin receptor, KLRG1, as one important candidate to study mechanistically. KLRG1 is an immunoreceptor tyrosine inhibitory motif domain-containing receptor that was originally characterized as an inhibitory receptor on natural killer (NK) cells, and has been used as a marker for terminally differentiated effector T cells.^{3–5} The ligands for KLRG1 are E-cadherins, N-cadherins and R-cadherins, which are highly expressed by solid tumors.^{6,7} A highly conserved region interacts with the C-type lectin-like domain of KLRG1, making this interaction a promising candidate for blockade to improve immune-mediated tumor control.

Recent studies have found that KLRG1 blockade, particularly in combination with PD-1 blockade, can decrease tumor burden in transplantable tumor cell mouse models.^{8,9} However, the mechanism behind KLRG1 blockade and a wider examination of the immune cell subsets that mediate this increased antitumor immune response remains to be further explored. Importantly, in vitro studies have shown that human KLRG1 is significantly more inhibitory than mouse KLRG1.¹⁰ The inhibitory function of human KLRG1 in the antitumor immune response may be underestimated in standard mouse syngeneic models. Thus, an in vivo model incorporating human KLRG1 is needed.

To investigate the functional role of KLRG1 in antitumor immunity in vivo, we generated a KLRG1 knockout (KO) (KLRG1^{-/-}) mouse. KLRG1^{-/-} mice showed increased spontaneous tumor control, supporting KLRG1 as a critical negative regulator of antitumor immunity. Knocking out N-cadherin from tumor cells resulted in a similar slowed tumor growth, suggesting that tumor cells are providing the functionally relevant ligand. In order to directly explore the function of human KLRG1, we generated a human KLRG1 knock-in (human KLRG1 KI) mouse model, and an anti-human KLRG1 blocking monoclonal antibody (mAb) (αKLRG1) was developed. Human KLRG1 KI mice showed accelerated tumor growth, consistent with the predicted stronger immune suppressive effect. αKLRG1 potentiated human T-cell activation in vitro, and αKLRG1 treatment of tumor-bearing human KLRG1 KI mice led to markedly reduced tumor growth. The enhanced antitumor effects were contributed to by CD8⁺ T cells, NK cells, and γδ T cells. Altogether, these data highlight the therapeutic potential of targeting human KLRG1 signaling to improve immune-mediated tumor control through multiple subsets of immune cells.

RESULTS

Mouse and melanoma patient T cells upregulate KLRG1 expression in response to checkpoint blockade treatment

We previously reported that a subset of tumor-infiltrating T cells expressed KLRG1, which was further augmented with checkpoint blockade.¹¹ To explore this dynamic further, we used the well-characterized B16.SIY model of melanoma and treatment with several immune checkpoint antibodies α-programmed death-ligand 1 (PD-L1), αCTLA-4, α4-1BB, and/or α-lymphocyte-activation gene 3 (LAG3) (figure 1A, online supplemental figure 1). Refractory tumors were harvested, infiltrating immune cells were isolated, and flow cytometry was performed to analyze levels of KLRG1 expression in tumor antigen-specific SIY⁺ CD8⁺ T cells. KLRG1 expression on SIY⁺ CD8⁺ T cells increased significantly in response to α4-1BB and multiple combination therapies. Our recent work indicating a role for host 4-1BB in anti-PD-1/PD-L1 efficacy suggests clinical relevance for the role of 4-1BB.¹² These results indicate that KLRG1 can be upregulated on tumor-infiltrating CD8⁺ T cells after initial checkpoint blockade therapy and suggest the hypothesis that KLRG1 might be an additional inhibitory receptor restraining the function of these cells.

To investigate the relevance of KLRG1 expression in human melanoma, we re-analyzed immune cell populations in a published single-cell RNA sequencing (scRNAseq) dataset from pretreatment and post-treatment tumor biopsies of patients with melanoma treated with αPD-1.¹³ Overall, αPD-1 non-responder patients had a higher frequency of KLRG1 expression across various T-cell populations (figure 1B). This included terminally differentiated effector memory CD8⁺ T cells and gamma delta (γδ) T cells. These results suggest that patients treated with anti-PD-1 have upregulated expression of KLRG1 on tumor-infiltrating T cells, which is enriched in non-responder patients. Analysis of effector and exhaustion genes that correlate with elevated KLRG1 levels in non-responding patients to anti-PD-1 revealed a significant increase in granzymes A, B, and K, along with increased PRF1 and IFNG expression (online supplemental table 1). Furthermore, elevated KLRG1 expression in non-responder patients correlated with higher expression of exhaustion markers (T cell immunoglobulin and mucin domain-containing molecule 3 (TIM3), T cell immunoglobulin and ITIM domain (TIGIT)) and significantly lower expression of T cell factor 1 (TCF-1), suggesting that KLRG1 expression is associated with a shift from a stem-like state toward a more highly cytolytic yet exhausted phenotype.

Host KLRG1 functions as a significant negative regulator of the antitumor immune response in vivo

To assess the potential functional relevance of KLRG1 on the antitumor immune response, we generated KLRG1 KO mice (KLRG1^{-/-}). KLRG1^{-/-} mice were healthy and showed no obvious phenotype of autoimmunity. B16.SIY transplantable tumor cells were subcutaneously injected

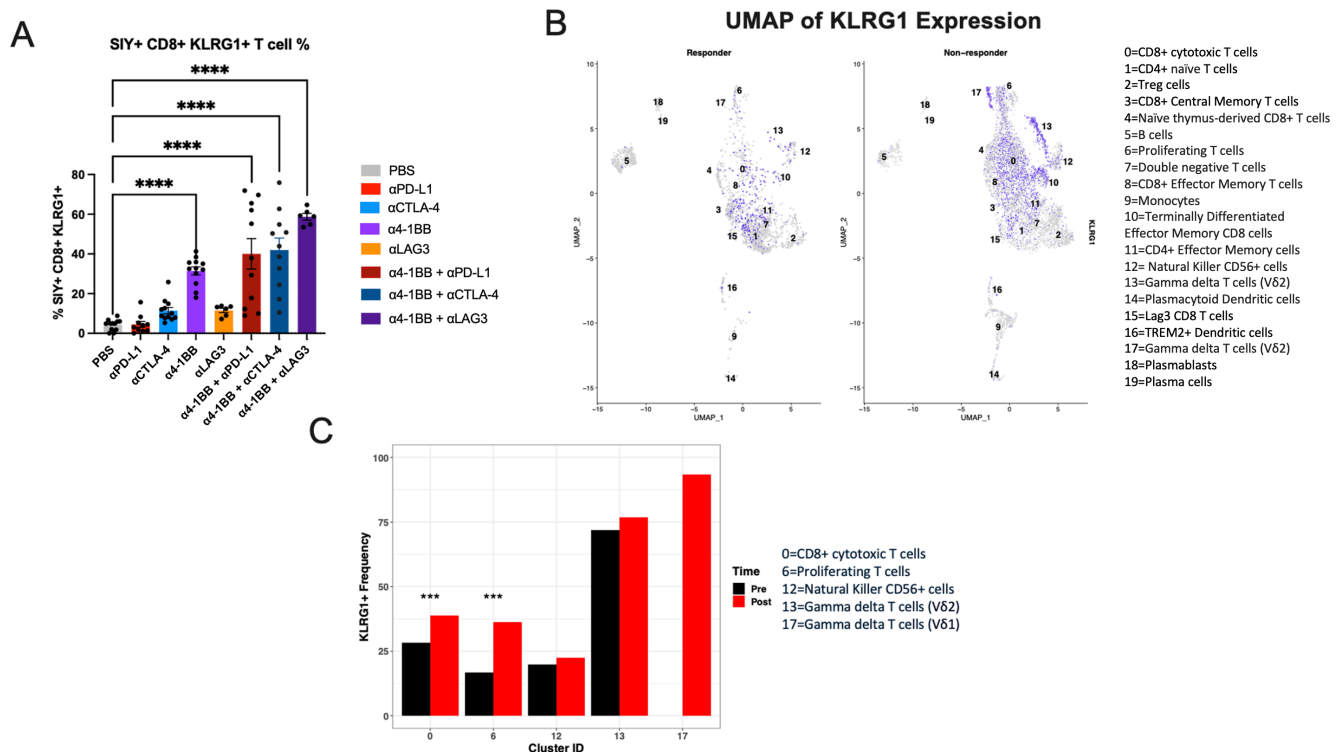


Figure 1 Higher KLRG1 expression in αPD-1 post-treatment non-responder patient with melanoma T-cell populations. (A) Summary of KLRG1 expression on SIY⁺CD8⁺ T cells in B16.SIY tumors following single or combination treatment with αPD-L1, αCTLA-4, α4-1BB, and/or αLAG3. Mice received antibody doses (100 μg each) intraperitoneally on days 7, 10, 13, and 16 after tumor inoculation. Tumors were analyzed on day 25 following single antibody treatment or day 35 following combination antibody treatment. Pooled data are plotted from two independent experiments with n≥6 per group. (B) UMAP plots of single-cell RNA sequencing data of CD45⁺ cells from 35 melanoma tumor biopsies of patients treated with αPD-1. UMAPs show cells that express KLRG1 in immune cells from responder (left) and non-responder (right) patient with melanoma tumor biopsy samples. (C) Frequency of KLRG1-expressing cells in specific immune cell clusters from patients with melanoma tumor biopsies pretreatment and post-treatment with αPD-1. Bar graph comparisons were analyzed by unpaired, two-tailed Student's t-test and plotted as mean±SEM. Significance is defined as *p<0.05, **p<0.01, ***p<0.001, ****p<0.0001. CTLA-4, cytotoxic T lymphocyte-associated molecule-4; PD-1, programmed death-1; PD-L1, programmed death-ligand 1; LAG3, lymphocyte-activation gene 3; UMAP, uniform manifold approximation and projection.

into KLRG1^{-/-} mice and KLRG1^{+/+} littermate control mice, and the tumor growth rate was assessed over time. KLRG1^{-/-} mice showed significantly diminished tumor growth compared with KLRG1^{+/+} mice, indicating it is likely a critical negative regulator of antitumor immunity (figure 2A). Successful loss of KLRG1 in CD8⁺ T cells was confirmed by flow cytometry (online supplemental figure 2). KLRG1^{-/-} mouse tumors had a significantly increased number of infiltrating SIY⁺ CD8⁺ T cells, NK cells, and γδ T cells compared with KLRG1^{+/+} mice (figure 2B–C, gating strategy shown in online supplemental figure 3). This immune phenotype was confirmed in a second transplantable tumor cell model, MC38.SIY (online supplemental figure 4). KLRG1^{-/-} mouse tumors also had an overall increase of CD45⁺ immune cells, CD4⁺ T cells, and monocytes (online supplemental figure 5). Furthermore, tumor CD8⁺ T cells from KLRG1^{-/-} mice had an increased frequency of central memory T cells compared with KLRG1^{+/+} mice (online supplemental figure 6). A significantly increased frequency of interferon (IFN)-γ-secreting SIY-specific T cells was also observed in the spleen

(figure 2D). These results indicate that the improved tumor control in KLRG1^{-/-} mice was associated with a broadly augmented endogenous immune response that includes antigen-specific CD8⁺ T cells, NK cells, and γδ T cells.

CD8⁺ T cells, NK cells, and γδ T cells functionally contribute to the augmented tumor control in KLRG1^{-/-} mice

A series of immune cell depletion experiments was done where either CD8⁺ T cells, NK cells, or γδ T cells were depleted by administration of specific antibodies 1 day before subcutaneous injection of B16.SIY tumor cells, and then again 7 and 14 days later. Successful depletion of each immune cell population was confirmed by flow cytometry of blood samples following antibody administration (online supplemental figure 7). Depletion of CD8⁺ T cells or NK cells from KLRG1^{-/-} mice resulted in a complete loss of immune-mediated tumor control, indicating both populations contribute to the antitumor phenotype (figure 3A–B). Furthermore, depletion of γδ T cells led to a partial decrease in tumor control (figure 3C).

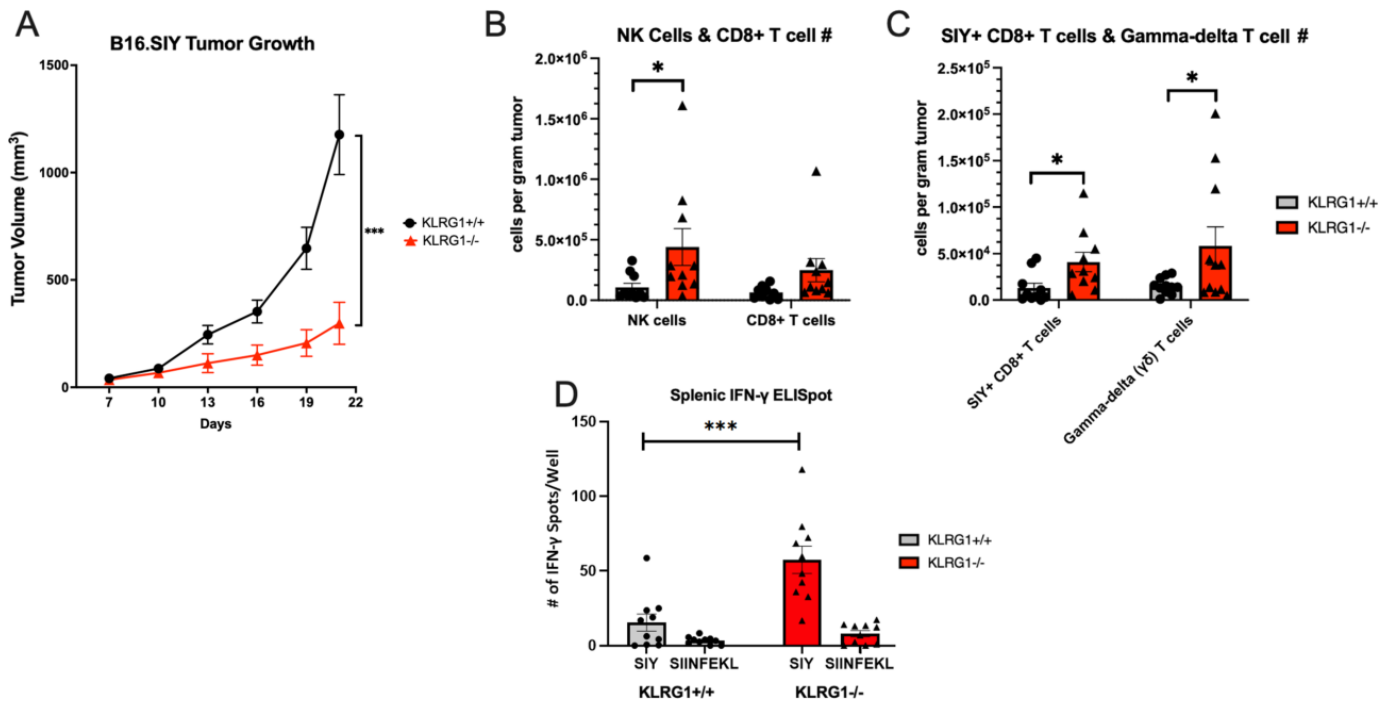


Figure 2 Loss of KLRG1 leads to increased tumor control and increased number of NK cells, antigen-specific CD8⁺ T cells, and γδ T cells in the tumor microenvironment. (A) Tumor outgrowth in KLRG1^{+/+} and KLRG1^{-/-} mice subcutaneously inoculated with B16.SIY tumors. Pooled data are plotted from two independent experiments with n=12 per group and plotted as mean±SEM. (B) Absolute numbers of immune cell subsets measured by flow cytometry staining from tumors 21 days after subcutaneous injection. (C) Frequency of splenic IFN-γ-secreting T cells in response to SIY and SIINFKL (negative control) measured by ELISpot. Splenocytes were isolated from mice on day 21 after tumor inoculation and cultured with 160nM SIY or for SIINFKL peptide for 24 hours. Pooled data are plotted from two independent experiments with n=10 per group. Tumor growth curves were analyzed by two-way ANOVA with Tukey's post-test and bar graph comparisons were analyzed by unpaired, two-tailed Student's t-test and plotted as mean±SEM. Significance is defined as *p<0.05, **p<0.01, ***p<0.001, ****p<0.0001. ANOVA, analysis of variance; ELISpot, enzyme-linked immunospot; IFN, interferon; NK, natural killer.

These data indicate that CD8⁺ T cells and NK cells are required for mediating the improved tumor control in KLRG1^{-/-} mice, along with a partial dependency on γδ T cells.

To determine if the enhanced tumor control in KLRG1 KO mice has a combinatorial effect with PD-1/PD-L1 blockade, tumor growth experiments were done with the administration of αPD-L1 Ab to KLRG1^{-/-} mice. Treating KLRG1^{-/-} B16.SIY tumor-bearing mice with αPD-L1 resulted in modest additional therapeutic benefit compared with tumor growth in KLRG1^{-/-} with vehicle treatment (figure 3D). However, the combination of anti-PD-L1 treatment and host KLRG1 deletion was not significantly more efficacious than anti-PD-L1 alone, in this setting where anti-PD-L1 is already highly effective.

Tumor cell-intrinsic N-cadherin expression mediates inhibition of immune response through KLRG1

An attractive feature of KLRG1 is that its defined ligands are E-cadherin and N-cadherin, which are broadly expressed by many cancer cell types, suggesting KLRG1 could mediate inhibition of activated T cells on direct interaction with tumor cells.^{14 15} However, cadherins also can be expressed by normal host cells, including immune cells.^{16 17} Through RNA-seq of the

B16 mouse melanoma cell line, we identified N-cadherin as the most highly expressed cadherin (online supplemental figure 8). We therefore generated N-cadherin KO B16.SIY cell lines (NCAD⁻) using CRISPR/Cas9 and rescued N-cadherin expression by transducing NCAD⁻ cells with an N-cadherin expression vector (NCAD restored) (figure 4B). KLRG1^{+/+} and KLRG1^{-/-} mice were subcutaneously injected with NCAD⁻, NCAD restored, or control empty vector-transduced B16.SIY cells and tumor growth was measured over time. B16.SIY cells lacking N-cadherin grew much more slowly in vivo compared with B16.SIY cells that expressed N-cadherin (figure 4C). Furthermore, tumor growth of B16.SIY cells with and without N-cadherin was equally reduced in KLRG1^{-/-} mice, supporting that N-cadherin and KLRG1 are in the same pathway. Reintroduction of N-cadherin restored tumor growth in KLRG1^{+/+} mice and yet growth was still reduced in KLRG1^{-/-} mice. Together, these data indicate that N-cadherin specifically expressed by tumor cells, and not N-cadherin elsewhere in the host, mediates the inhibition of the antitumor immune response through KLRG1 signaling. Analysis of infiltrating immune cells by flow cytometry in endpoint tumors showed that KLRG1^{+/+} mice with B16.SIY cells

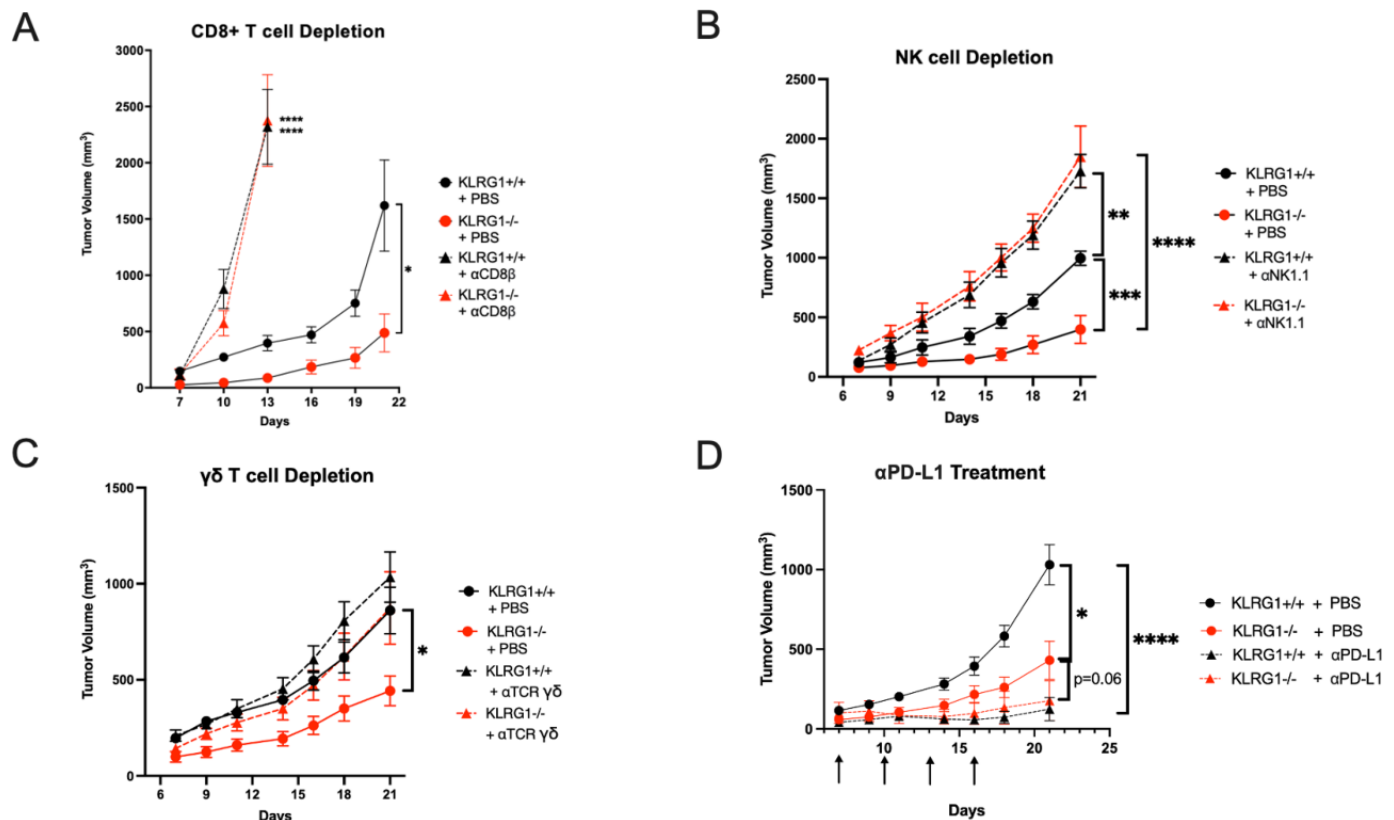


Figure 3 Enhanced tumor control in KLRG1^{-/-} mice is dependent on CD8⁺ T cells and NK cells. B16.SIY tumor growth curves of (A) CD8⁺ T cell-depleted mice, (B) NK cell-depleted mice or (C) gamma delta (γδ) T cell-depleted mice. KLRG1^{+/+} and KLRG1^{-/-} received 200 μg of αCD8α (clone 53–5.8), αNK1.1 (clone PK136), or αTCR γδ (clone UC7-13D5) through IP injection 1 day before tumor cell injection then once per week. Pooled data are plotted from two independent experiments with n=10 per group. (D) Tumor growth curves of KLRG1^{+/+} and KLRG1^{-/-} treated intraperitoneally with 100 μg of αPD-L1 or vehicle (PBS) on the days indicated by the arrows. Pooled data are plotted from two independent experiments with n=10 per group. Tumor growth curves were analyzed by two-way ANOVA with Tukey's post-test and plotted as mean±SEM. Significance is defined as *p<0.05, **p<0.01, ***p<0.001, ****p<0.0001. ANOVA, analysis of variance; NK, natural killer; PBS, phosphate-buffered saline; PD-L1, programmed death-ligand 1; IP, intraperitoneal.

that lacked N-cadherin had significantly increased numbers of SIY⁺CD8⁺T cells and NK cells in the TME, which were reduced to baseline when N-cadherin expression was restored (figure 4D). No significant differences in KLRG1 expression were observed on CD8⁺ T cells and NK cells between NCAD⁺ and NCAD⁻ tumors (online supplemental figure 9). To confirm loss of N-cadherin-mediated signaling increases T-cell killing, we conducted cytotoxicity experiments using activated CD8⁺ T cells isolated from KLRG1^{+/+} or KLRG1^{-/-} mice with MC57.SIY tumors. T cells were primed with MC57.SIY then tested against B16.SIY to evaluate the specificity of the immune response towards the SIY antigen. CD8⁺ T cells from KLRG1^{-/-} mice had significantly more killing of B16.SIY tumor cells compared with CD8⁺ T cells from KLRG1^{+/+} mice (figure 4E). Furthermore, there was greater killing by KLRG1^{+/+} CD8 T cells against N-cadherin-deleted B16.SIY tumors compared with B16.SIY tumors that still expressed N-cadherin. KLRG1^{-/-} CD8⁺ T cells showed high amounts of B16.SIY killing whether or not N-cadherin was deleted, with no statistically

significant difference between killing NCAD⁺ and NCAD⁻ tumors by KLRG1^{-/-} CD8⁺ T cells. Together, these results demonstrate that N-cadherin/KLRG1 interactions inhibit T cell-mediated cytotoxicity.

Generation of human KLRG1 knock-in mice and an anti-human αKLRG1 blocking antibody

Human KLRG1 is more inhibitory than murine KLRG1 because of superior formation of dimers.¹⁰ To determine the immune regulatory effect of human KLRG1 in vivo, we developed a human KLRG1 KI mouse model. We confirmed that human KLRG1 binds to mouse N-cadherin through flow cytometry and incubation of human KLRG1-expressing cells with mouse N-cadherin (online supplemental figure 10). Humanized KLRG1 KI mice had significantly increased B16.SIY tumor cell growth compared with wild-type (WT) mice, supporting that there may be stronger immunosuppressive effects of human KLRG1 compared with mouse KLRG1 (figure 5A).

To study the therapeutic potential of blocking human KLRG1 signaling, we developed a

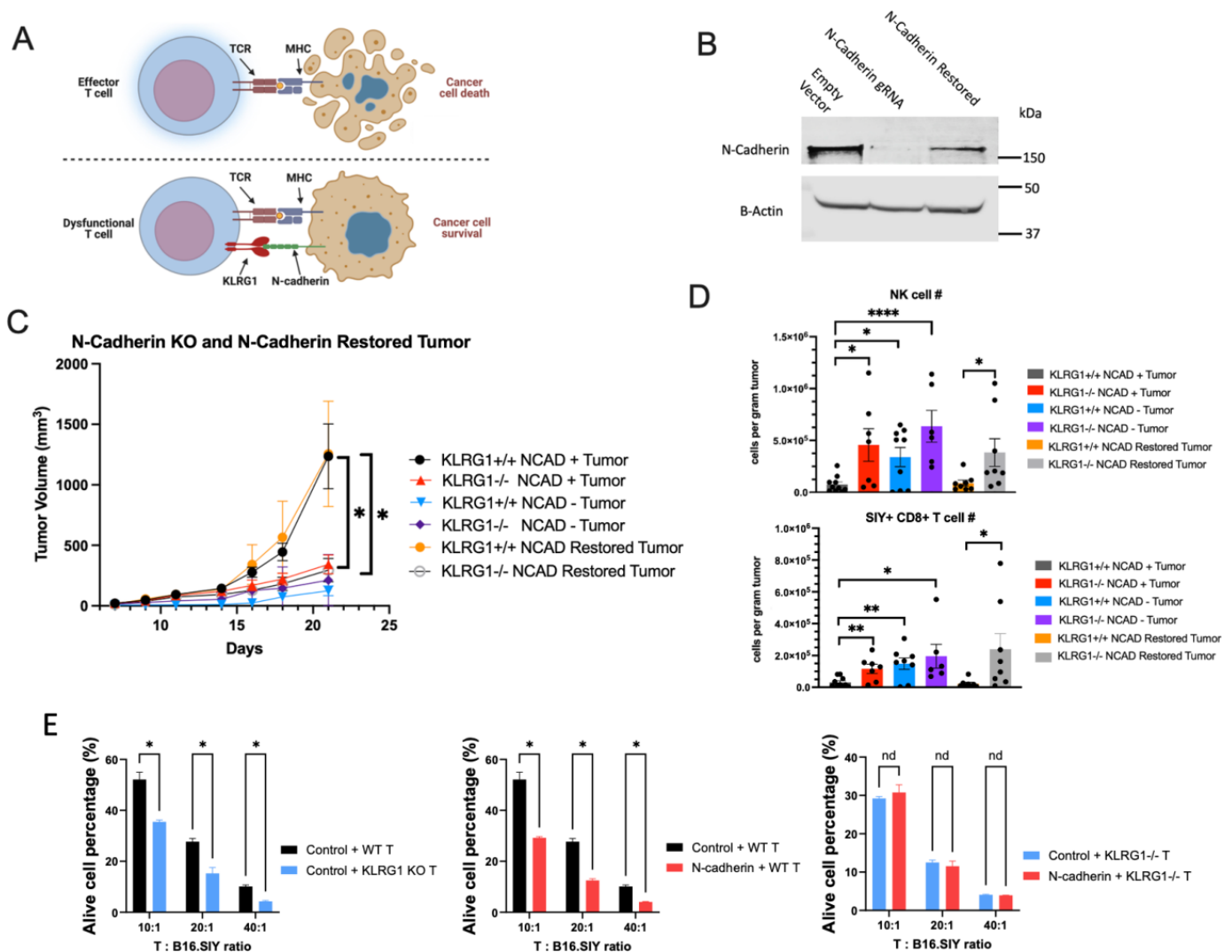


Figure 4 Loss of host KLRG1 and/or tumor N-cadherin decreases B16.SIY tumor growth. (A) Diagram of the interaction between KLRG1 and tumor-expressed N-cadherin causing loss of effector T-cell function. (B) Western blot of N-cadherin from B16.SIY cell lysates of the same subclone transfected with empty vector, N-cadherin CRISPR/Cas9 KO plasmid, or N-cadherin expression rescued using N-cadherin overexpression vector. (C) Tumor outgrowth in KLRG1^{+/+} or KLRG1^{-/-} mice subcutaneously inoculated with control empty vector transfected B16.SIY cells, B16.SIY cells transfected with N-cadherin CRISPR/Cas9 KO plasmid, or rescued N-cadherin expressing B16.SIY cells. Pooled data are plotted from two independent experiments with $n=10$ per group and plotted as mean $\pm$ SEM. (D) Absolute numbers of immune cell subsets measured by flow cytometry staining from tumors 21 days after subcutaneous injection. (E) Viability of B16.SIY cells with or without N-cadherin depletion after 16 hours of culturing with activated CD8⁺ T cells from KLRG1^{+/+} or KLRG1^{-/-} mice that had MC57.SIY tumors. Pooled data are plotted from two independent experiments with $n=6-10$ per group. Tumor growth curves were analyzed by two-way ANOVA with Tukey's post-test and bar graph comparisons were analyzed by unpaired, two-tailed Student's t-test and plotted as mean $\pm$ SEM. Significance is defined as * $p<0.05$, ** $p<0.01$, *** $p<0.001$, **** $p<0.0001$. ANOVA, analysis of variance; KO, knockout; MHC, major histocompatibility complex; TCR, T-cell receptor; WT, wild-type.

humanized IgG4-antibody which targets human KLRG1 (α KLRG1). The IgG4 isotype was selected to reduce Fc-mediated antibody-dependent cellular cytotoxicity (ADCC) activity. Confirmation that α KLRG1 can block KLRG1 receptor-ligand binding was done using labeled KLRG1-tetramers and E-cadherin overexpressing K562 cells (figure 6A). α KLRG1 significantly blocked KLRG1 tetramer binding to E-cadherin overexpressing cells compared with human IgG4-antibody isotype control, with a binding affinity (half-maximal inhibitory concentration (IC_{50})) of

1.13 nM. Additionally, α KLRG1 demonstrated capability of binding to CD8⁺ T cells (half-maximal effective concentration (EC_{50})=2.3 nM) isolated from human peripheral blood mononuclear cells (PBMCs) (figure 6B). To test the functionality of α KLRG1 on human CD8⁺ T cells, we used co-culturing of KLRG1⁺ CD8⁺ T cells from healthy donors with artificial antigen-presenting cells (APCs) that expressed T-cell receptor (TCR)-activating ligands along with E-cadherin. Addition of α KLRG1 augmented secretion of

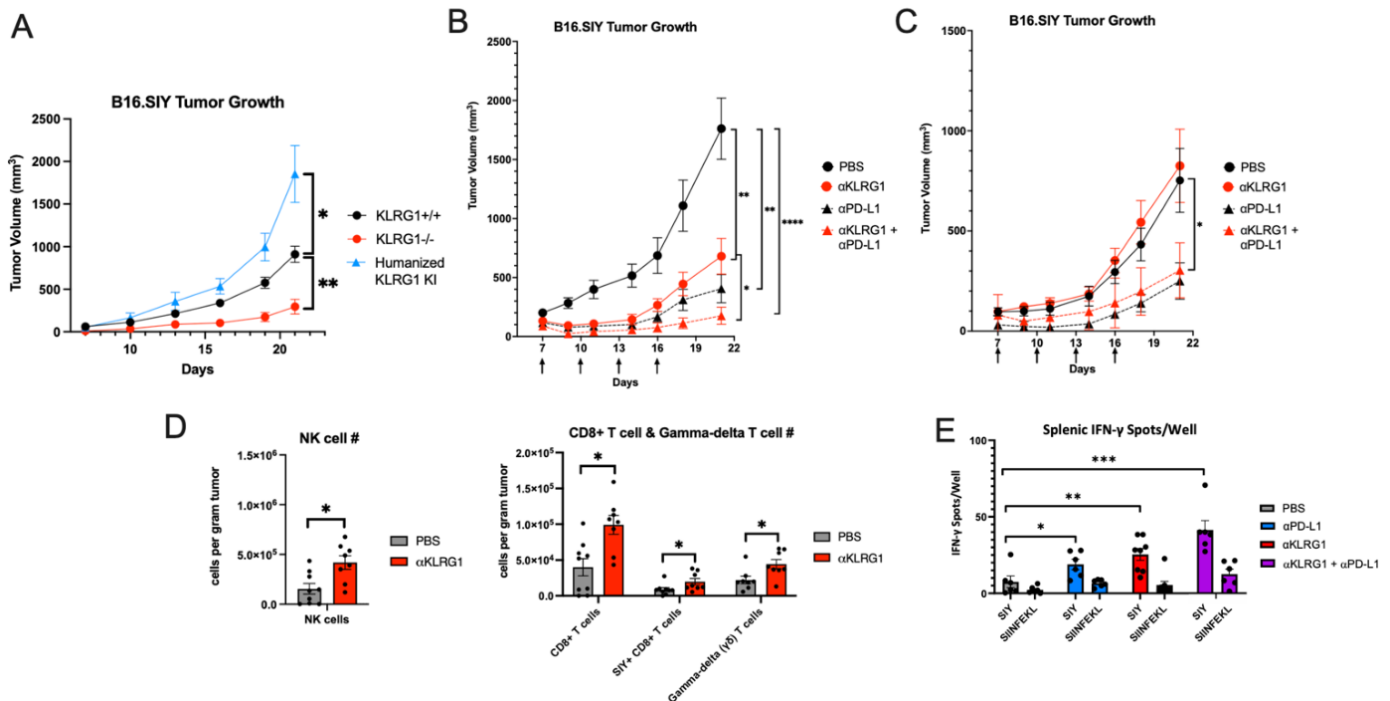


Figure 5 Significant decrease of tumor growth in humanized KLRG1 knock-in mice after αKLRG1 treatment. (A) Tumor outgrowth in KLRG1^{+/+} and KLRG1^{-/-} mice subcutaneously inoculated with B16.SIY tumors. n=5 per group and plotted as mean±SEM. B16.SIY tumor outgrowth of (B) humanized KLRG1 mice and (C) WT mice treated with PBS, αPD-L1, αKLRG1, or αPD-L1 and αKLRG1 combination treatment. Mice received 100 μg of each antibody intraperitoneally on days 7, 10, 13, and 16 after tumor inoculation. Pooled data are plotted from two independent experiments with n=5–10 per group and plotted as mean±SEM. Tumor growth curves were analyzed by two-way ANOVA with Tukey's post-test. (D) Absolute numbers of immune cell subsets measured by flow cytometry staining from tumors 12 days after subcutaneous injection from humanized KLRG1 KI mice treated with PBS or αKLRG1. Pooled data are plotted from two independent experiments with n≥8 per group. (E) Frequency of splenic IFN-γ-secreting T cells in response to SIY and SIINFKL (negative control) measured by ELISpot. Splenocytes were isolated from mice on day 12 after tumor inoculation and cultured with 160 nM SIY or for SIINFKL peptide for 24 hours. Pooled data are plotted from two independent experiments with n=6–8 per group. Bar graph comparisons were analyzed by unpaired, two-tailed Student's t-test and plotted as mean±SEM. Significance is defined as *p<0.05, **p<0.01, ***p<0.001, ****p<0.0001. ANOVA, analysis of variance; ELISpot, enzyme-linked immunospot; IFN, interferon; KI, knock-in; NK, natural killer; PBS, phosphate-buffered saline; PD-L1, programmed death-ligand 1; WT, wild-type.

the pro-inflammatory cytokines IFN-γ (EC₅₀=0.35 nM) and tumor necrosis factor (TNF)-α (EC₅₀=0.37 nM) (figure 6C–D). These data demonstrate that αKLRG1 blocks the KLRG1-cadherin interaction and potentiates the activation of human CD8⁺ T cells in vitro.

Human KLRG1 KI mouse model reveals marked therapeutic efficacy of αKLRG1 treatment, which is dependent on CD8⁺ T cells and NK cells

The human KLRG1 KI mouse model provided a means to test the therapeutic potential of anti-human KLRG1 mAb in an in vivo setting. Treatment of humanized KLRG1 KI mice with the human αKLRG1 mAb resulted in significantly decreased tumor growth compared with phosphate-buffered saline (PBS) (figure 6B). Additionally, combination treatment of αKLRG1 and αPD-L1 resulted in further reduction of tumor growth compared with treatment with each mAb alone. However, the combination treatment was not significantly more efficacious than αPD-L1 treatment alone, in this mouse model where αPD-L1 is already highly effective (figure 5B). Anti-human KLRG1 had no therapeutic effect on tumor

growth in WT mice, which only have mouse KLRG1, confirming its specificity for human KLRG1 (figure 5C).

To assess the potential mechanisms of the improved tumor control with αKLRG1 treatment, tumor-infiltrating immune cells were analyzed using flow cytometry. Human KLRG1 KI mice treated with anti-human KLRG1 mAb had significantly increased numbers of NK cells, SIY⁺ CD8⁺ T cells, and γδ T cells in the TME compared with vehicle-treated mice, similar to the phenotype in KLRG1^{-/-} mice (figure 5D). These tumors also had markedly increased numbers of CD4⁺ T cells, plasmacytoid dendritic cells (pDCs), and conventional type 1 dendritic cells (cDC1) cells (online supplemental figure 11). Similar to KLRG1^{-/-} mice, tumor CD8⁺ T cells from mice treated with αKLRG1 had a significantly increased frequency of central memory T cells compared with PBS-treated mice (online supplemental figure 6). Furthermore, splenocytes from B16.SIY tumor-bearing humanized KLRG1 KI mice treated with αKLRG1 showed a significantly increased frequency of IFN-γ secreting T cells as assessed by IFN-γ enzyme-linked immunospot (ELISpot) (figure 5E). The combination

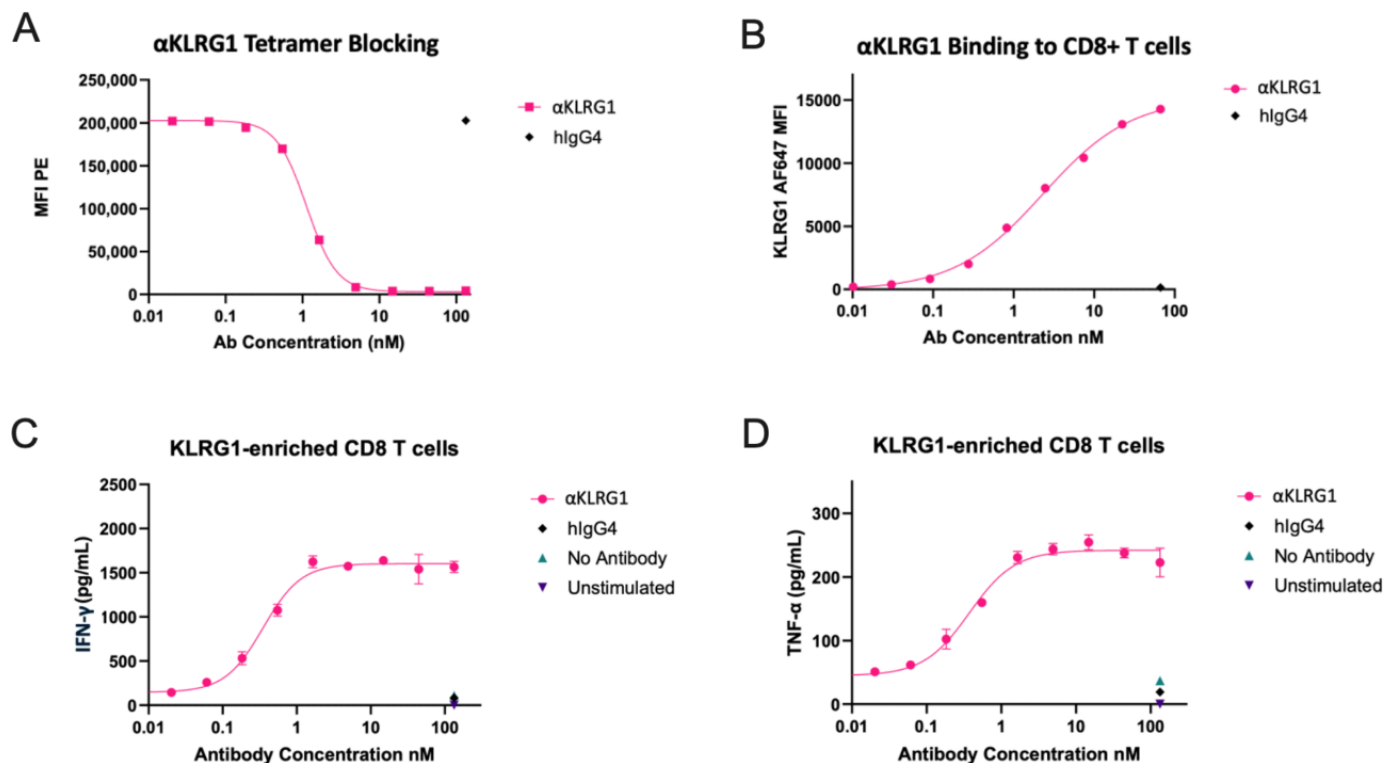


Figure 6 Human α KLRG1 mAb blocks KLRG1 signaling and enhances human T-cell activation in vitro. K562 cells overexpressing human E-cadherin were preincubated with KLRG1-tetramer-PE followed by incubation with increasing antibody concentrations ranging from 0.01 to 100 nM of α KLRG1 mAb or control hlgG4. Flow cytometry was used to measure the ability of the α KLRG1 mAb to block the interaction between human KLRG1 and E-cadherin (A). Human PBMCs were stained with α KLRG1 and human IgG4 antibodies directly conjugated to Alex Fluor 647, then flow cytometry was used to measure the binding affinity of α KLRG1 to CD8⁺ T cells (B). Human CD8⁺ T cells FACS sorted to enrich for KLRG1⁺CD8⁺ T cells were added to TCR activator CHOK1 cells, which overexpress E-cadherin, and preincubated with increasing antibody concentrations ranging from 0.01 to 100 nM of α KLRG1 mAb, hlgG4, or no antibody or unstimulated controls. Secreted (C) IFN- γ and (D) TNF- α levels in supernatants after 3 days were measured by ELISA. IFN, interferon; mAb, monoclonal antibody; PBMC, peripheral blood mononuclear cell; PE, phycoerythrin; TCR, T-cell receptor; TNF, tumor necrosis factor; FACS, fluorescence-activated cell sorting; MFI, mean fluorescence intensity.

of α KLRG1 and α PD-L1 significantly increased the frequency of splenic IFN- γ secreting T cells in humanized KLRG1 mice compared with either treatment alone, showing a combinatorial effect of both treatments on T-cell functionality. These results support that α KLRG1 can improve tumor control in vivo through increasing the activation of multiple immune cells.

To interrogate which immune cell populations were contributing to the improved tumor control in humanized KLRG1 KI mice after treatment with α KLRG1, a series of immune cell depletion experiments were done. Depleting either CD8⁺ T cells or NK cells leads to a complete loss of increased tumor control from α KLRG1 treatment (figure 7A–B). Depletion of $\gamma\delta$ T cells in humanized KLRG1 KI mice treated with α KLRG1 led to a modest loss of tumor control (figure 7C). These data support that the α KLRG1 mAb increases the antitumor immune response through both CD8⁺ T cells and NK cells, with a partial contribution of $\gamma\delta$ T cells.

KLRG1 expression correlates with the presence of a T cell-inflamed tumor microenvironment across multiple human cancer types, which also express cadherins

Our data suggest that targeting KLRG1 could be attractive to advance to the clinical setting. We therefore pursued a broader investigation into the relationship between KLRG1 expression and a T cell-inflamed TME across a range of human cancers. We examined the bulk RNA-seq data from 32 solid tumor cancers from TCGA and categorized each tumor type into inflamed, intermediate, and non-inflamed cohorts using transcripts that are consistently correlated with a T cell-inflamed gene signature¹⁸ (figure 8A). Expression of KLRG1 was significantly higher in inflamed tumor cohorts across most cancer types, except in ovarian and glioblastoma cancers (online supplemental table 2). Specifically, breast, head and neck, and colon cancers showed the most significant increase in KLRG1 expression in inflamed tumors compared with non-inflamed tumors. Furthermore, most solid tumors showed high expression levels of E-cadherin and 15 different solid tumors showed elevated levels of N-cadherin expression (figure 8B). Bulk RNA-seq analysis

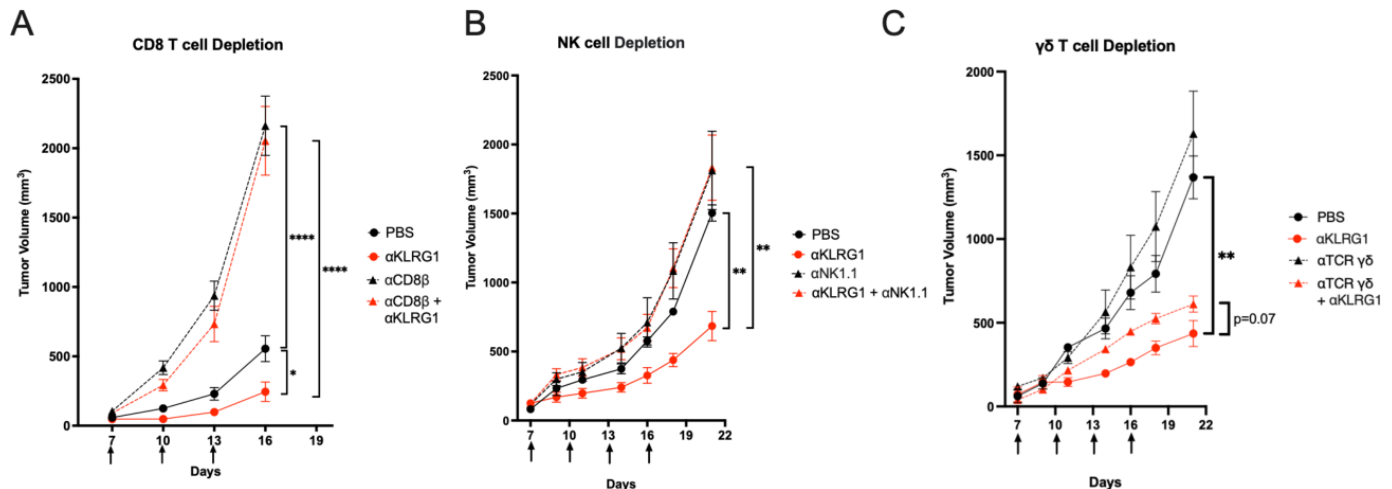


Figure 7 Enhanced tumor control of α KLRG1 treatment is dependent on CD8⁺ T cells and NK cells. B16.SIY tumor growth curves of WT or humanized KLRG1 KI mice treated with PBS or α KLRG1 along with depletion of (A) CD8⁺ T cells, (B) NK cells or (C) gamma delta ($\gamma\delta$) T cells. WT or humanized KLRG1 KI mice received 200 μ g of α CD8 α (clone 53–5.8), α NK1.1 (clone PK136), or α TCR γ/δ (clone UC7-13D5) through IP injection 1 day before tumor cell injection then once per week. Pooled data are plotted from two independent experiments with $n=10$ per group. Tumor growth curves were analyzed by two-way ANOVA with Tukey's post-test and plotted as mean $\pm$ SEM. Significance is defined as * $p<0.05$, ** $p<0.01$, *** $p<0.001$, **** $p<0.0001$. ANOVA, analysis of variance; KI, knock-in; NK, natural killer; PBS, phosphate-buffered saline; TCR, T-cell receptor; WT, wild-type; IP, intraperitoneal.

of tumors from α PD-1-treated patients with melanoma revealed significantly higher E-cadherin and N-cadherin expression in non-responders compared with responders (online supplemental figure 12). These results show that in most solid tumors, there is high KLRG1 expression in T cell-inflamed tumors and an abundance of ligand to mediate inhibitory signaling from KLRG1. These cancer types could be considered for α KLRG1 therapy as clinical development is pursued.

DISCUSSION

We identified KLRG1 as a promising therapeutic target based on our previous screen of molecules expressed on the surface of mouse dysfunctional CD8⁺ T cells in the TME. In mice, treatment with known immune checkpoint Abs led to further KLRG1 upregulation. In the human context, the frequency of KLRG1-expressing T cells was greatly increased in patients with melanoma tumors treated with α PD-1, which was even more pronounced in non-responders. Our results suggest that KLRG1 may represent a dominant inhibitory receptor allowing continued immune suppression even when PD-1/PD-L1 interactions are blocked.

Cadherins such as E-cadherin and N-cadherin are known to play key roles in cancer progression, including the epithelial-mesenchymal transition and promoting the metastatic capacity of tumor cells.^{14 15 19} The pervasive expression of cadherins in solid tumors supports the potential of therapeutically targeting the KLRG1-cadherin interaction. Cadherins are expressed by other cells in the TME, such as N-cadherin expression by endothelial cells and E-cadherin by dendritic cells.^{16 20} Previous studies have shown that KLRG1 blocking antibodies in combination with anti-PD-1 can decrease tumor growth

with E-cadherin and N-cadherin expressing cell lines.^{8 9}

Our use of N-cadherin KO tumor cell lines expands on these findings and supports that N-cadherin specifically on tumor cells is the dominant ligand-mediated inhibition of antitumor immunity in this model.

Our human KLRG1 KI model is the first to show that tumor growth is accelerated compared with endogenous murine KLRG1 in vivo. This observation suggests that KLRG1 blockade in humans could provide more therapeutic benefit than what blocking KLRG1 signaling in standard mouse models can indicate. Despite that augmented tumor growth, α KLRG1 treatment had remarkable therapeutic efficacy, through a mechanism contributed by CD8⁺ T cells, NK cells, and $\gamma\delta$ T cells.

KLRG1 expression was observed across solid tumors in the TCGA dataset having a T cell-inflamed TME. Interestingly, the T cell-inflamed tumors with the highest KLRG1 expression levels are typically non-responsive to current checkpoint blockade therapies, including prostate, pancreatic, and breast cancers.^{21–23} In addition, E-cadherin was expressed in almost all solid tumors and some tumors concurrently expressed elevated levels of N-cadherin. Together, these observations generate the hypothesis that KLRG1 may represent a major immune inhibitory receptor in these challenging tumor types, making α KLRG1 mAb treatment an attractive candidate for clinical translation.

MATERIALS AND METHODS

Study design

Mouse experiments were carried out at least two times, in an independent way, with $n\geq 5$. Higher mouse numbers were used in experiments with a broader value distribution. Sample sizes were chosen based on previously

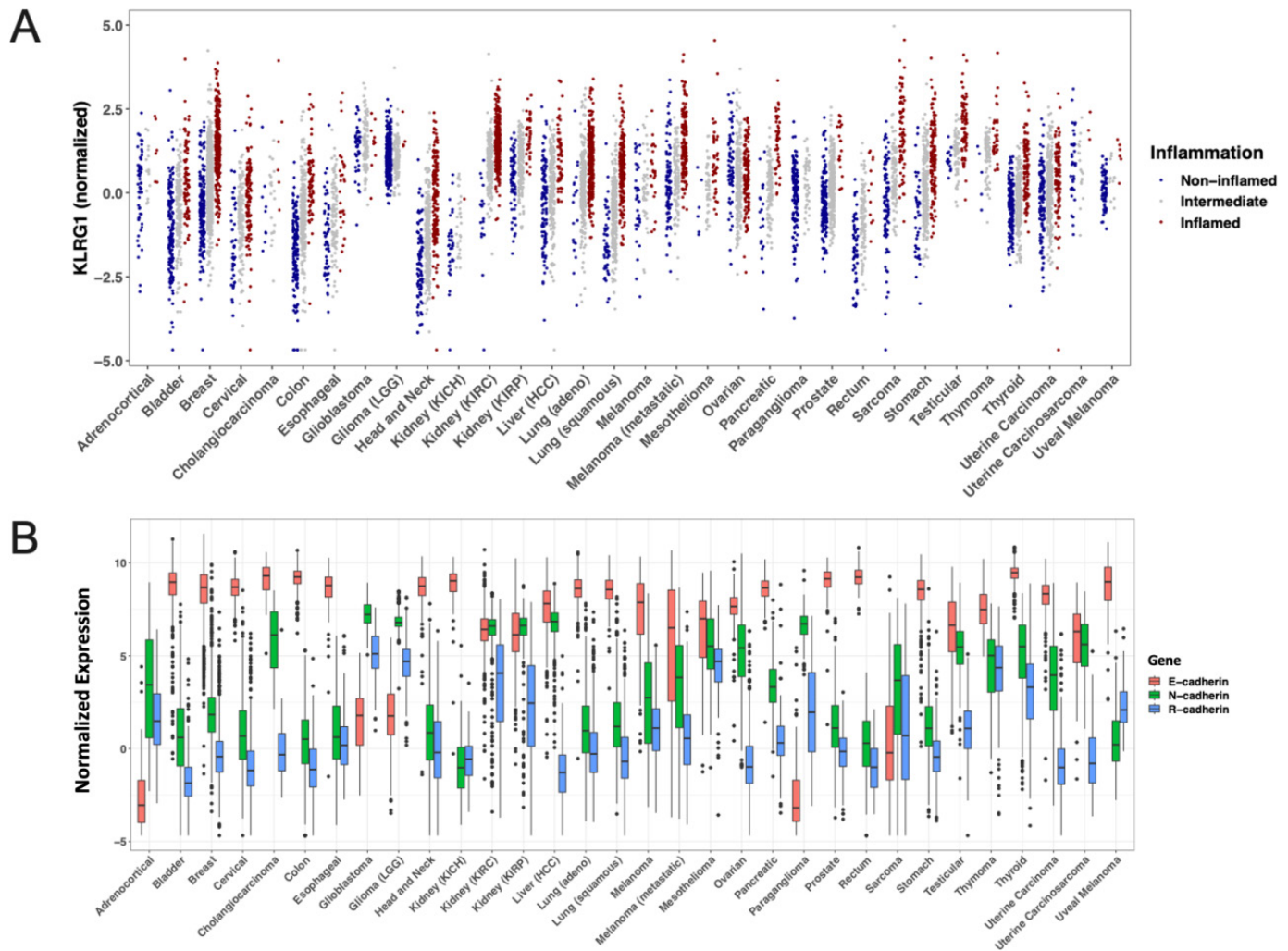


Figure 8 KLRG1 and cadherin expression in the inflamed tumor microenvironment across human cancer types. (A) Distribution of normalized KLRG1 transcriptional expression levels in non-inflamed (blue), intermediate (gray), and inflamed (red) tumors of each TCGA cancer type. Each dot represents a single sample. (B) Normalized transcriptional expression levels of E-cadherin (red), N-cadherin (green), and R-cadherin (blue) across each TCGA cancer type. KICH, kidney chromophobe; LGG, low-grade glioma; KIRP, kidney renal papillary cell carcinoma; KIRC, kidney renal clear cell carcinoma; HCC, hepatocellular carcinoma; TCGA, The Cancer Genome Atlas.

published studies. Mice that failed to develop tumors from the beginning were excluded from the analysis. No data outliers were excluded. Mice were randomized after tumor establishment for treatments. Tumor growth experiments had a predetermined endpoint based on tumor size or tumors were collected based on expectation for maximal lymphocyte response. Whenever possible, investigators were blinded during experiments to group allocation and when assessing outcomes. For each experiment, the number of replicates, statistical tests used, and P values are reported in the figure legends. All animal maintenance and experiments were conducted following the guidelines established by the National Institute of Health Animal Care and studied under Institutional Animal Care and Use Committee (IACUC)-approved protocols. For human biological samples, all samples were collected with prior patient consent.

Single-cell RNA analysis

Pre-processed count data and patient metadata from Sade-Feldman *et al* were obtained from Gene Expression Omnibus (GEO) (GSE120575) and converted to a Seurat (V.4.3.0) object in R (V.4.2.2).²⁴ We subsetted the data to only patients treated with anti-PD-1 and to genes shared with the 10X Visium v1 probes, re-scaled the data with Seurat's ScaleData function, and followed a standard Seurat workflow to build a principal component analysis (PCA) and uniform manifold approximation and projection (UMAP) of the data. We used the function FindClusters to identify clusters at a range of resolutions and proceeded with downstream analysis using a resolution of 1.0. We then identified cluster markers using the function FindConservedMarkers, using the treatment metadata as a grouping variable. Putative cell types were annotated manually from the top markers for each cluster. The fully processed data were then used to compare KLRG1

expression in each cluster among responders and non-responders. For a subset of CD8⁺ T-cell clusters (0, 6, 12, 13, and 17), we compared the frequency of KLRG1⁺ cells between pretreatment and post-treatment samples using a χ^2 proportion test. P values were corrected for multiple comparisons with the Benjamini-Hochberg method.

TCGA analysis

All available bulk RNA data and corresponding patient metadata from all TCGA projects were downloaded from the National Cancer Institute (NCI) Genomic Data Portal (accessed November 2023). Samples were restricted to primary tumors, as well as metastases in the case of melanoma (from TCGA-skin cutaneous melanoma (SKCM)). Samples from lymphoma (TCGA-diffuse large B-cell lymphoma (DLBC)) were excluded from the analysis. The final dataset included 10,247 samples.

Using the edgeR (V.3.40.2) package in R, we filtered out any genes with fewer than five counts in 90% of samples.²⁵ We then converted the values to log-counts-per-million (log2-cpm), and upper quartile-normalized the data with the calcNormFactors function. These values were converted to row Z-scores to ensure a consistent expression distribution when estimating gene signatures across samples. We then estimated a 13-gene T-cell inflammation signature using the following genes: *CD8A*, *CCL2*, *CCL3*, *CCL4*, *CXCL9*, *CXCL10*, *ICOS*, *GZMK*, *IRF1*, *HLA-DMA*, *HLA-DMB*, *HLA-DOA*, and *HLA-DOB*. The signature was estimated as the median expression of these genes in each sample across the Z-scores calculated above. Samples were considered “Non-inflamed” if they fell below the lower quartile of inflammation score distribution, “Inflamed” if their score fell in the upper quartile, and “Intermediate” otherwise. We then compared normalized KLRG1 expression (using upper quartile-normalized values) between non-inflamed, intermediate, and inflamed samples, separated by each TCGA project. Within each project, we compared KLRG1 expression between inflamed and non-inflamed samples using a one-sided t-test with the alternative hypothesis that expression would be higher in the inflamed samples. We corrected for multiple comparisons using the Benjamini-Hochberg method. Additionally, we compared expression of E-cadherin (*CDH1*), N-cadherin (*CDH2*), and R-cadherin (*CDH4*) across TCGA projects.

Bulk RNA-seq analysis

RNA and biopsy samples were obtained using previously described methods.²⁶ All tumor specimens were collected prior to anti-PD-1-based therapy treatment. Patients were required to have documented clinical outcomes categorized as complete response, partial response, stable disease, or progressive disease.

Sequencing data were aligned to the GENCODE primary assembly sequence (release 29) using Kallisto and tximport. Raw counts were filtered to include only non-duplicated, protein-coding genes and processed using edgeR. After applying trimmed mean of M-values

(TMM)-normalization with calcNormFactors, the counts were converted to log2-cpm. These processed data were used to compare the expression levels of *CDH1*, *CDH2*, and *CDH4*.

15 of the RNA samples were previously reported in a study by Cron *et al.*²⁶, and the data are available from the GEO (GSE285619). Data for 41 additional samples were added under the new accession GSE319884.²⁶

Mouse models

All experiments were done with mice on the C57BL/6 background. Both the KLRG1^{-/-} mice and the humanized KLRG1 KI mice were generated for our laboratory by Cyagen. KLRG1^{-/-} mice were generated by inserting a targeting vector with the c.26_30delCACTA point loss mutation into exon 1 of the endogenous locus of the *Klrk1* gene. To generate the humanized KLRG1 KI mouse model, the region of mouse *Klrk1* from aa.57 plus partial intron 2 was replaced by the Human KLRG1 CDS (aa.60-195)-3'UTR of mouse *Klrk1*-rBG pA cassette, using a targeting vector for the endogenous locus of the mouse *Klrk1* gene. C57BL/6 mice were purchased from Taconic (RRID:MGI:2164831). All mice used in experiments were between the age range of 6–12 weeks and were age-matched, litter-matched, and sex-matched within experiments.

Syngeneic tumor models

Murine cancer cell lines syngeneic to the C57BL/6 background were used for tumor growth experiments. B16F10 (RRID:CVCL_0159) (ATCC) cells were engineered to express the model antigen SIYRYGL fused in-frame with DsRed. Similarly, the murine colorectal cancer cell line MC38 was transduced to express SIY in conjunction with green fluorescent protein (GFP). Using the respective fluorophores, SIY expression was routinely monitored over time with flow cytometry analysis and flow cytometric sorting was done periodically to ensure uniform expression and achieve consistent immunogenicity. HEK293T cells (RRID:CVCL_0063) were purchased from ATCC. For inoculation of tumor cells, murine tumor cell lines were harvested and resuspended in PBS at a concentration of 20 million cells per mL. Each mouse was injected subcutaneously with 100 μ L of cell suspension (2 million cells). Tumors were measured with digital calipers starting on day 7 post-inoculation and tumor volume was calculated using the following formula: width $\times$ height $\times$ length=volume (cm³).

Antibody administration

For antibody treatments, mice received intraperitoneally 100 μ g of anti-PD-L1 (10F.9G2: Bio X Cell) and/or 100 μ g anti-CTLA-4 (BE0131: Bio X Cell) and/or 100 μ g of anti-4-1BB (Bio X Cell; LOB12.3) and/or 100 μ g of anti-LAG-3 (Bio X Cell; C9B7W) and/or 100 μ g of anti-human KLRG1 depending on the experiment on days 7, 10, 13, and 16 after tumor engraftment. For immune cell depletion experiments, mice received 200 μ g of either anti-CD8b

(YTS 169.4: Bio X Cell), anti-NK1.1 (PK136: Bio X Cell), anti-mouse TCR γ/δ (UC7-13D5: Bio X Cell), or anti-CD4 (GK1.5: Bio X Cell) 1 day before tumor engraftment then weekly until endpoint. The depletion of relevant immune cell populations was confirmed by flow cytometry analysis of blood 3 days after antibody administration and at the endpoint of tumor growth.

TIL isolation

Tumors and spleens were harvested from mice at the indicated time points. Spleens were minced on ice and passed through 40 μ M filters then red blood cell (RBC)-lysed following the manufacturer's protocol (BD, 555899). For tumor-infiltrating lymphocytes (TIL) isolation, tumors were minced on ice, then digested by incubating with an enzyme mix containing collagenase (1 mg/mL, Sigma-Aldrich: C5138), DNase (200 μ g/mL, Sigma-Aldrich: D5025), and hyaluronidase (100 μ g/mL, Sigma-Aldrich: H6254) for 30 min at 37°C while shaking. This tumor suspension was filtered through a 40 μ m filter, then washed with PBS. TILs were further enriched by Ficoll-Paque (Fisher, 17144003) density gradient centrifugation.

Flow cytometry

Cell suspensions were incubated with Fc Block (BioLegend, 101302) and SIY-pentamer (ProImmune, F1803-2B—150 test R-PE) in Brilliant Stain Buffer Plus (BD, 566385) at 1:3 dilution in PBS for 30 min. Antibody master mix was added to these samples and the samples were incubated again for 30 min on ice. Antibodies against the following molecules were used: CD3 (17A2, AF700), CD8a (53–6.7, PerCP eF710), NK1.1 (PK136, APC Cy7), TCR γ/δ (GL3, BV421), KLRG1 (2F1, BV605), PD-1 (J43, BUV395), CD86 (PO3, BUV496), CD69 (H12F3, BUV563), Ly6G (1A8, BUV661), CD103 (2E7, BV421), CD19 (6D5, BV510), CD11b (M1/70, BV570), CD206 (C068C2, BV605), CD317 (927, BV650), Ki67 (B56, BV711), CD4 (GK1.5, Spark 685), Foxp3 (FJK-16s, APC), Lag3 (C9B7W, BUV805), Ly6C (HK1.4, BV785), CD44 (IM7, FITC), CD45 (30-F11, AF532), F4/80 (BM8, PerCP/Cy5.5), CD80 (16–10A1, PE/Dazzle 594), CD62L (MEL-14, PE/Cy7), CD11c (N418, AF 647), I-A/I-E (M5/114.15.2, Pacific Blue). Following washing, cells were fixed and permeabilized using the FoxP3 staining kit (BD Biosciences) according to manufacturer's instructions combined with the fixable live/death discrimination dye (Zombie NIR, BioLegend). Cells were then stained with the intracellular master mix and incubated overnight at 4°C. All spectral flow cytometry was done on 5L Cytex Aurora machines and analyzed using FlowJo software (RRID:SCR_008520).

IFN- γ ELISpot

ELISpot assays were used to determine the frequency of IFN- γ producing CD8⁺ T cells. ELISpot plates (Millipore MultiScreen IP Plates Catalog #: MAIPS4510, Millipore, Danvers Massachusetts, USA) were coated with anti-mouse IFN- γ ELISpot capture antibody (100 μ L/

well; BD Biosciences 51–2525KZ) and stored overnight at 4°C. ELISpot plates were washed and blocked with Dulbecco's Modified Eagle Medium (DMEM) supplemented with 10% fetal calf serum (FCS) for 2 hours at room temperature. Splenocytes from tumor-challenged mice were harvested and plated at 10⁶ cells/well. Cells were stimulated with either 160 nM SIY peptide, 160 nM SIINFEKL peptide as a negative control, or a 50 ng/mL phorbol 12-myristate 13-acetate (PMA) and 500 ng/mL ionomycin mixture as a positive control. Plates were then stored at 37°C in an 8% CO₂ incubator overnight. The next day, the plates were washed and coated with anti-mouse IFN- γ ELISpot Detection Antibody (BD Biosciences 51–1818KA) for 2 hours at room temperature. Plates were then washed and coated with HRP-conjugated streptavidin (BD Biosciences #554066) for 1 hour at room temperature. Plates were washed, developed with the addition of AEC substrate, then dried overnight. ELISpot images were captured using CTL ImmunoSpot Analyzer. The spot number was counted using ImmunoSpot Software (RRID:SCR_011082).

Cell line genetic manipulation

The N-cadherin sgRNA expression vector was constructed with the forward (5'-caccACGCGCGCAGGCGGCGAGGGgt-3') and reverse (5'-taaacCCCTCGCCGCTCGCGCGGT-3') oligonucleotides mixed at 10 mM each in 10 mM Tris-HCl (pH8.0) and 5 mM MgCl₂ in a total volume of 100 μ L. This mixture was incubated at 95°C for 5 min then cooled to room temperature. The duplex oligonucleotide was cloned into the BbsI site of pKLV-U6sgRNA(BbsI)-PGKpuro2ABFP (RRID:Ad-dgene_50946). Transient transfection of the N-cadherin sgRNA expression vector or empty control vector along with the Cas9 GFP (Applied Biological Material) vector into B16.SIY cells was performed using Lipofectamine 3000 reagent (Thermo Fisher Scientific) according to the manufacturer's protocol. Single cells were sorted on GFP (Cas9) and BFP (sgRNA) expression using an FACSaria II cell sorter (BD Biosciences). After the expansion of single-cell clones, KO of N-cadherin expression in each clone was assessed using western blot.

To rescue N-cadherin expression, control vector and N-cadherin overexpression vector were obtained from VectorBuilder. The N-cadherin KO B16.SIY cells were stably transduced with lentivirus expressing N-cadherin and sorted based on GFP (overexpression vector) fluorescence. Expression of N-cadherin was confirmed by western blot.

T-cell killing assay

MC57.SIY cells were injected subcutaneously into WT or *Klhl1* KO littermate control mice. On day 7 after tumor injection, spleens were harvested from tumor-bearing mice, and CD8⁺ T cells were purified by negative selection using a CD8 T Cell Purification Kit (Miltenyi Biotec). Purified CD8⁺ T cells were counted and seeded in 6-well plates at a density of 5 × 10⁶ cells/mL. Cells were stimulated with

anti-CD3 (1 µg/mL), anti-CD28 (1 µg/mL), and recombinant interleukin-2 (50 µ/mL).

After 48 hours of stimulation, SIY-specific CD8⁺ T cells were stained and analyzed by flow cytometry to determine the frequency of SIY⁺ CD8⁺ T cells. Activated CD8⁺ T cells were then co-cultured with B16.SIY tumor cells. After 16 hours, B16.SIY cells were stained with a live/dead viability dye and analyzed by flow cytometry. Tumor cell viability was calculated as the number of live B16.SIY cells normalized to the live cell number in medium-only control wells.

Western blotting

For immunoblot analysis, cells were harvested and lysed on ice for 10 min with radioimmunoprecipitation assay buffer. Lysates were then centrifuged to remove cell debris, and protein concentrations were determined by bicinchoninic acid protein assay (Pierce, Thermo Fisher #23225). Samples were then heat-treated at 95°C in sodium dodecyl sulfate (SDS) loading buffer. Lysates were separated on SDS polyacrylamide gels then transferred to polyvinylidene (PVDF) membranes (Millipore, Billerica, Massachusetts, USA). Membranes were blocked with 5% non-fat milk, then immunoblotted with primary antibodies during overnight incubations at 4°C. The membranes were washed three times with tris-buffered saline with tween-20 (TBST), then incubated for 1 hour with appropriate secondary antibodies at room temperature. For probing, the antibodies used were: N-cadherin (Abcam, ab76011) and β-Actin (1:1,000; rabbit monoclonal, Cell Signaling Technology, #4970S). Membranes were visualized using ImageQuant LAS 4000 (General Electric Company).

Anti-KLRG1 monoclonal antibody production

The anti-KLRG1 antibody clone 1D21 v1 is a proprietary antibody developed by Pyxis Oncology (Patent PCT/US2024/034971). Briefly, C57Bl/6 WT mice were immunized with human KLRG1 extracellular domain-hFc fusion protein. Serum antibody titer after immunization was assessed by ELISA using hKLRG1.hFc and flow cytometry using human KLRG1 overexpressing cell line. Mice that showed the highest titers against human KLRG1 were euthanized for hybridoma production. Hybridomas were further screened by flow cytometry, ELISA, and bio-layer interferometry, to measure the binding affinity and specificity of the antibody to human-KLRG1. Hybridomas producing high binding affinity antibodies to human KLRG1 were subcloned and sequenced. Then, the variable regions of the lead clone 1D21 were further humanized, deimmunized, and fused to the human IgG4 constant region. Finally, the lead variant 1D21 v1 antibody with human IgG4 backbone was selected as lead antibody by preserving its high binding affinity and specificity to human-KLRG1. For in vivo studies, the original lead clone 1D21 in mouse IgG1 backbone with Fc silent mutation (D265A) was used.

Human KLRG1 tetramer and E-cadherin blocking assay

Biotinylated human KLRG1 extracellular domain was mixed with streptavidin-phycoerythrin (PE) to produce the KLRG1-tetramer-PE. K562 cells overexpressing human E-cadherin were resuspended at 1×10^6 cells/mL in complete Iscove's Modified Dulbecco's Medium (IMDM) (Thermo Fisher). Cells were first blocked with Fc block (BioLegend) and anti-CD32a antibody (Bio X cell) for 10 min on ice. Human KLRG1 tetramer-PE was added to the cells at a final concentration of 620 nM. Next, 100 µL of the cells were plated into 96 well U-bottom plates. The 1D21 v1 antibody was resuspended in complete IMDM medium (3X, serial dilution 1/4) and added to the cells at a final concentration from 133 nM to 0.02 nM. The Human IgG4 isotype was added at a final concentration of 133 nM. Cells were incubated for 2 hours on ice, washed three times with complete IMDM, and resuspended in IMDM. The ability of the 1D21 v1 antibody to block the interaction between human KLRG1 and E-cadherin was analyzed by flow cytometry.

Binding assay to human CD8⁺ T lymphocytes

Human healthy PBMCs were thawed and resuspended in 10 mL PBS completed with 10% heat-inactivated serum. Cells were blocked with 1:50 Human TruStain Fc α (BioLegend) for 20 min on ice. 1D21 v1 and human IgG4 antibodies were directly conjugated to Alex Fluor 647 (AF647) with the APEX antibody conjugation kit (Invitrogen) according to the manufacturer's protocol. Antibody concentrations were measured by NanoDrop and antibodies were resuspended at a final concentration from 66.6 nM to 0.01 nM (serial dilution 1/3). CD3, CD8, CD14, CD4 (BioLegend), CD19 (Invitrogen), and CD56 (BD Biosciences) were also added at 1:100 dilution to the cell gating. Cells were stained for 30 min on ice, washed two times with PBS completed with 10% heat-inactivated serum and 1D21 v1 binding affinity to CD8⁺ T lymphocyte was analyzed by flow cytometry.

IFN- γ and TNF- α secretion assay

TCR activator CHOK1 cells overexpressing human E-cadherin were treated with mitomycin C for 1 hour before plating 2×10^5 cells on a 96-well plates overnight. Human CD8⁺ T lymphocytes were purchased from stem cell. CD8⁺ CCR7-negative T lymphocytes were FACS sorted to enrich for KLRG1⁺CD8⁺ T cells. 5×10^4 CCR7⁻CD8⁺ T cells were added to the TCR activator CHOK1 cells and incubated with 1D21 v1 antibody at a final concentration from 133 nM to 0.01 nM (serial dilution 1/3) or 133 nM of the human IgG4 isotype for 3 days in a 37°C and 5% CO₂ humidified incubator. Cell supernatants were collected and the amount of secreted IFN- γ and TNF- α protein was measured by ELISA using the MabTech kits according to the manufacturer's protocol to assess the activation of KLRG1⁺CD8⁺ T cells.

Statistical analysis

Following ARRIVE guidelines, all statistical analyses were done using GraphPad Prism software (GraphPad). For

all statistical analysis, data are shown as means±SEM. Tumor growth curves were analyzed by two-way analysis of variance with Tukey's multiple comparisons test. Bar graph comparisons were analyzed using unpaired, two-tailed Student's t-test (parametric). Raw FCS files were unmixed in Cytek's SpectroFlo software. Flow cytometry data counts and percentages were calculated using FlowJo software. P values<0.05 were considered statistically significant and represented as: *p<0.05, **p<0.01, ***p<0.001, ****p<0.0001. We used the ARRIVE reporting guideline to draft this manuscript, and the ARRIVE reporting checklist when editing, included as supplemental attachment.^{27 28}

Contributors TAJ and TG designed the project. TAJ, SS, KY, SL and MI designed and performed the experiments. TAJ, KY, MI, SL, NS and JP analyzed and interpreted the data. JWS performed computational analysis. NS and JWS contributed to making figures and writing the methods section. TAJ and TG wrote the manuscript. TG is the guarantor of this study. All authors reviewed and edited the manuscript.

Competing interests The authors KY, MI, NS, and JP are employees and shareholders in Pyxis Oncology. TG is a founder of Pyxis Oncology and a member of its scientific advisory board and a shareholder in Pyxis Oncology. The other authors declare that they have no competing interests.

Patient consent for publication Not applicable.

Ethics approval Informed consent was secured for patients with melanoma, with data acquisition conducted under University of Chicago Biological Sciences Division Institutional Review Board protocols 14-0735 and 15-0837. Research procedures adhered to ethical standards, including the US Common Rule, the Belmont Report, and the Declaration of Helsinki. Participants gave informed consent to participate in the study before taking part.

Provenance and peer review Not commissioned; externally peer reviewed.

Data availability statement Data are available upon reasonable request.

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REFERENCES

- Topalian SL, Sznol M, McDermott DF, et al. Survival, durable tumor remission, and long-term safety in patients with advanced melanoma receiving nivolumab. *J Clin Oncol* 2014;32:1020–30.
- Leach DR, Krummel MF, Allison JP. Enhancement of antitumor immunity by CTLA-4 blockade. *Science* 1996;271:1734–6.
- Robbins SH, Nguyen KB, Takahashi N, et al. Cutting edge: inhibitory functions of the killer cell lectin-like receptor G1 molecule during the activation of mouse NK cells. *J Immunol* 2002;168:2585–9.
- Voehringer D, Koschella M, Pircher H. Lack of proliferative capacity of human effector and memory T cells expressing killer cell lectinlike receptor G1 (KLRG1). *Blood* 2002;100:3698–702.
- Voehringer D, Blaser C, Brawand P, et al. Viral infections induce abundant numbers of senescent CD8 T cells. *J Immunol* 2001;167:4838–43.
- Ito M, Maruyama T, Saito N, et al. Killer cell lectin-like receptor G1 binds three members of the classical cadherin family to inhibit NK cell cytotoxicity. *J Exp Med* 2006;203:289–95.
- Tessmer MS, Fugere C, Stevenaert F, et al. KLRG1 binds cadherins and preferentially associates with SHIP-1. *Int Immunol* 2007;19:391–400.
- Greenberg SA, Kong SW, Thompson E, et al. Co-inhibitory T cell receptor KLRG1: human cancer expression and efficacy of neutralization in murine cancer models. *Oncotarget* 2019;10:1399–406.
- Tata A, Dodard G, Fugère C, et al. Combination blockade of KLRG1 and PD-1 promotes immune control of local and disseminated cancers. *Oncoimmunology* 2021;10:1933808.
- Hofmann M, Schweier O, Pircher H. Different inhibitory capacities of human and mouse KLRG1 are linked to distinct disulfide-mediated oligomerizations. *Eur J Immunol* 2012;42:2484–90.
- Williams JB, Horton BL, Zheng Y, et al. The EGR2 targets LAG-3 and 4-1BB describe and regulate dysfunctional antigen-specific CD8+ T cells in the tumor microenvironment. *J Exp Med* 2017;214:381–400.
- Ziblat A, Horton BL, Higgs EF, et al. Batf3+ DCs and the 4-1BB/4-1BBL axis are required at the effector phase in the tumor microenvironment for PD-1/PD-L1 blockade efficacy. *Cell Rep* 2024;43:114141.
- Sade-Feldman M, Yizhak K, Bjorgaard SL, et al. Defining T Cell States Associated with Response to Checkpoint Immunotherapy in Melanoma. *Cell* 2018;175:998–1013.
- Yu W, Yang L, Li T, et al. Cadherin Signaling in Cancer: Its Functions and Role as a Therapeutic Target. *Front Oncol* 2019;9:989.
- Thiery JP. Epithelial-mesenchymal transitions in tumour progression. *Nat Rev Cancer* 2002;2:442–54.
- Cao ZQ, Wang Z, Leng P. Aberrant N-cadherin expression in cancer. *Biomed Pharmacother* 2019;118:109320.
- Van den Bossche J, Malissen B, Mantovani A, et al. Regulation and function of the E-cadherin/catenin complex in cells of the monocyte-macrophage lineage and DCs. *Blood* 2012;119:1623–33.
- Spranger S, Luke JJ, Bao R, et al. Density of immunogenic antigens does not explain the presence or absence of the T-cell-inflamed tumor microenvironment in melanoma. *Proc Natl Acad Sci U S A* 2016;113:E7759–68.
- Cavallaro U, Christofori G. Cell adhesion and signalling by cadherins and Ig-CAMs in cancer. *Nat Rev Cancer* 2004;4:118–32.
- Padovan E, Terracciano L, Certa U, et al. Interferon stimulated gene 15 constitutively produced by melanoma cells induces e-cadherin expression on human dendritic cells. *Cancer Res* 2002;62:3453–8.
- Galon J, Bruni D. Approaches to treat immune hot, altered and cold tumours with combination immunotherapies. *Nat Rev Drug Discov* 2019;18:197–218.
- Ho WJ, Jaffee EM, Zheng L. The tumour microenvironment in pancreatic cancer – clinical challenges and opportunities. *Nat Rev Clin Oncol* 2020;17:527–40.
- Semiglazov V, Tseluiko A, Kudaybergenova A, et al. Immunology and immunotherapy in breast cancer. *Cancer Biol Med* 2022;19:609–18.
- Hao Y, Hao S, Andersen-Nissen E, et al. Integrated analysis of multimodal single-cell data. *Cell* 2021;184:3573–87.
- Robinson MD, McCarthy DJ, Smyth GK. edgeR: a Bioconductor package for differential expression analysis of digital gene expression data. *Bioinformatics* 2010;26:139–40.
- Cron KR, Sivan A, Aquino-Michaels K, et al. PKCδ Germline Variants and Genetic Deletion in Mice Augment Antitumor Immunity through Regulation of Myeloid Cells. *Cancer Immunol Res* 2025;13:547–59.
- Percie du Sert N, Hurst V, Ahluwalia A, et al. The ARRIVE guidelines 2.0: Updated guidelines for reporting animal research. *PLoS Biol* 2020;18:e3000410.
- Sert NP, Hurst V, Ahluwalia A, et al. The arrive reporting checklist. In: Harwood J, Albury C, de BJ, et al, eds. *THE EQUATOR network reporting guideline platform [Internet]. The UK EQUATOR Centre.* 2025.